INTERNSHIP REPORT

Project 8

Global Supply Chain & Logistics Performance

Name : Naveen Kumar

Track : Operations Analytics

Platform : Google Colab

Dataset : DataCo Supply Chain Dataset

Tools Used

Python

Google Colab

Pandas

NumPy

Matplotlib

Seaborn

GitHub

Internship Project Report

Contents

1 Project Overview

The objective of this project is to analyze the DataCo Supply Chain Dataset using Python in Google Colab.

The project focuses on understanding logistics operations, shipping performance, customer behavior, sales trends and supply chain efficiency.

The entire project was divided into four notebooks.

- Notebook 01 : Data Loading
- Notebook 02 : Data Understanding
- Notebook 03 : Data Cleaning
- Notebook 04 : Exploratory Data Analysis

2 Dataset Description

The project uses the DataCo Supply Chain Dataset.

The dataset contains information related to

- Customer Details
- Orders
- Products
- Sales
- Profit
- Shipping Mode
- Delivery Status
- Market
- Country
- Region

The dataset contains approximately 180,000 records and 53 attributes which are used for supply chain analytics.

3 Work Completed

3.1 Notebook 01 : Data Loading

The following tasks were completed.

- Imported required Python libraries
- Uploaded archive.zip into Google Colab
- Extracted dataset
- Loaded CSV using Pandas
- Verified dataset
- Checked dataset dimensions
- Created backup dataset

3.2 Notebook 02 : Data Understanding

The following tasks were completed.

- Explored dataset structure
- Displayed first and last records
- Checked feature names
- Identified numerical and categorical columns
- Generated statistical summary
- Checked missing values
- Created distribution graphs

3.3 Notebook 03 : Data Cleaning

Completed activities

- Identified missing values
- Filled missing values

- Removed duplicate records
- Detected outliers
- Removed outliers using IQR method
- Saved cleaned dataset

3.4 Notebook 04 : Exploratory Data Analysis

Performed the following analysis

- Sales Distribution
- Profit Distribution
- Shipping Mode Analysis
- Market Analysis
- Product Category Analysis
- Country-wise Analysis
- Correlation Heatmap
- Business Insights

4 Graphs Generated

The following visualizations were created in Google Colab.

- Sales Distribution Histogram
- Profit Distribution Histogram
- Shipping Mode Bar Graph
- Product Category Bar Graph
- Market Distribution Pie Chart
- Delivery Status Pie Chart
- Country-wise Orders Bar Graph
- Correlation Heatmap

5 Key Findings

1. The dataset contains customer, order and logistics information.
2. Sales are generated from multiple international markets.
3. Standard shipping mode is the most frequently used shipping method.
4. Some product categories generate higher sales than others.
5. Profit varies significantly across different orders.
6. Certain countries contribute higher order volumes.
7. Delivery performance differs across shipping modes.
8. Market-wise demand shows noticeable variation.
9. Data cleaning improved overall dataset quality.
10. Exploratory Data Analysis revealed meaningful business patterns useful for supply chain optimization.

6 Challenges Faced

- Understanding dataset attributes
- Handling missing values
- Removing duplicate records
- Detecting outliers
- Performing data visualization
- Understanding logistics terminology

7 Conclusion

The Global Supply Chain and Logistics Performance project was successfully implemented using Google Colab and Python.

The dataset was loaded, explored, cleaned and analyzed successfully.

Various visualizations and business insights were generated that help understand sales performance, logistics efficiency, customer behaviour and shipping operations.

The project demonstrates the importance of data analytics in improving supply chain performance and supporting business decision making.

8 Future Scope

Future enhancements include

- Sales Prediction
- Demand Forecasting
- Shipment Delay Prediction
- Inventory Optimization
- Interactive Dashboard using Power BI
- Machine Learning Models

9 References

1. DataCo Supply Chain Dataset
2. Google Colab Documentation
3. Python Documentation
4. Pandas Documentation
5. NumPy Documentation
6. Matplotlib Documentation
7. Seaborn Documentation

******* End of Report *******